Sherif Hossam

Software Engineer (Unity)

+201061740212 ·sherif.hossam.k@gmail.com ·https://www.linkedin.com/in/sherif-hossam-90273a114 ·Cairo, Egypt ·Project Demos

SUMMARY

Unity/C# engineer with a decade of experience building real-time 3D, VR, and AR systems — from a production BIM viewer's UI and headless server tooling to AI-driven virtual avatars, architectural visualization pipelines, and multiplayer 3D platforms. Consistent focus on performance: reducing build size, optimizing rendering budgets, and keeping complex real-time environments running smoothly on real hardware. Now expanding into AI-integrated products, building on hands-on speech-to-text/text-to-speech and conversational-avatar experience.

SKILLS

C · C# · Unity Pro · Optimization · Docker · GitLab CI · Build Tooling · Headless/Server-side Rendering · UI Architecture · VR/3D Simulation

EXPERIENCE

Software Engineer — PlanRadar **Dec 2021 – Present**
Cairo, Egypt

- Own the UI layer (NewUi) of a production Unity-based BIM viewer as top author (~1,150 commits, 55% of that subsystem's history) — shipped across Web, iOS, Android, and UWP.
- Cut mesh memory by up to 92% on the heaviest models (3.56 GB → 272.5 MB) via a GPU-instanced batched-loading rewrite, and cut load time on the heaviest model from 5.56s to ~3s.
- Reduced build size ~25% (e.g. 246.2 MB → 185.6 MB) by switching the IL2CPP compiler configuration and removing unused packages, plus a separate iOS-specific build-size audit.
- Built headless Linux/Docker tooling (xvfb) to run the same viewer server-side, generating automated storey snapshots, pin-position images, and model memory reports, plus a server-side pipeline to split large BIM models into per-storey files for faster loading.
- Resolved 500+ Jira issues and authored technical documentation on build size, headless server rendering, and loading strategy.

Software Engineer — BE-STEAM **Mar 2020 – Dec 2021**
Cairo, Egypt

- Built a Unity-based 3D robot programming simulator for WRO (World Robot Olympiad) competitions, letting students program a virtual LEGO EV3/Spike Prime robot without physical hardware.
- Built a visual block-coding system that compiles drag-and-drop blocks into Python-style script, mirroring LEGO SPIKE Prime's real editor.
- Implemented full WRO 2023/2024 mission sets (Elementary/Junior/Senior) with simulated sensors, live telemetry, and scoring, plus the surrounding app shell (login, menus, robot customization, competition timer).

Software Engineer (Contract) — INTDV **Aug 2021 – Oct 2021**
Cairo, Egypt

- Part-time contract taken alongside primary Software Engineer role at BE-STEAM.
- Built an interactive AI chatbot avatar in Unity: converted user speech to text, sent it to a backend server for a response, then converted the returned text into synthesized speech (IBM Watson Text-to-Speech) with synced facial and mouth animation.
- Generated and rigged the 3D avatar using iClone (Reallusion), with real-time lip-sync driven by the SALSA plugin for Unity.
- Owned the interaction pipeline end-to-end — the real-time link between conversation text and believable avatar speech and animation.

Software Engineer — Elibre DMCC **Oct 2017 – May 2021**
Cairo, Egypt

- Led the flagship VR/AR pipeline: converted 2D architectural drawings into fully furnished, lit 3D scenes with automatic furniture placement — optimized to run in real time in VR at render quality comparable to offline renderers like 3ds Max.
- Built a dynamic space-fitting engine that computes which furniture fits a room's real dimensions and auto-lays it out, keeping generated scenes light enough for smooth VR framerates.
- Optimized real-time rendering and scene performance across the VR/AR product line (asset and lighting budgets, scene complexity) to keep large generated interiors running smoothly on headset hardware.
- Built a HoloLens AR application for shared, multi-user real-time walkthroughs and presentations of 3D building models.
- Also built a mobile AR indoor-navigation app for shopping malls, an in-house AR document scanner, and the underlying 2D-to-3D "Line Editor" and manual furnishing tools.

Software Engineer (Contract) — Arabtesting **Jul 2019 – Oct 2019**
Cairo, Egypt

- Part-time contract taken alongside primary Software Engineer role at Elibre DMCC.
- Built a VR application for therapeutic and learning support: exposure-therapy scenarios to help treat anxiety and specific phobias — including acrophobia (fear of heights) and claustrophobia (fear of enclosed spaces) — plus support tools for people with dyslexia.

Senior Computer Engineer — Arab Builders **Dec 2016 – Oct 2017**
Giza, Egypt

- Built "Simulation of City" — a real-time multiplayer 3D virtual shopping environment where multiple users could browse and shop together simultaneously.
- Built the multiplayer server from scratch in Node.js, integrated with Unity via a socket API to synchronize all connected users in the shared 3D world in real time.
- Built a companion seller application for merchants to list items in the 3D world, and the buyer-side shopping flow — browsing, selecting, and adding 3D-represented items to a cart.

Computer Engineer — Tronixs Egypt **Sep 2016 – Feb 2017**
Egypt

- Built a Virtual Drilling Rig Simulator (VDRS): VR projection system with multiple camera views, fully integrated with rig site equipment.
- Implemented hardware-accurate controls — driller's chair with dual joysticks, touch-screen well-control and mud-tank monitoring systems.
- Built an instructor station for monitoring trainee performance, running exercises, and recording/retrieving sessions.

Software Engineer (Freelance) — Freelancer.com **May 2013 – Oct 2016**
Cairo, Egypt

- Delivered freelance game development projects in Unity for international clients, including educational mini-games and advertising/promotional games (advergames).

EDUCATION

Cairo University **2013 – 2014**
Pre-Master, Computer Engineering

Misr Institute for Engineering & Technology **2008 – 2013**
Bachelor of Computer Engineering and Systems

LANGUAGES

Arabic (Native or Bilingual) English (Full Professional)

CERTIFICATIONS & HONORS

ROV Competition 2012 ROV Competition 2013